

Exercise 2.1 — Likelihood of Markov-Chain Histories

Recursive Macroeconomic Theory

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Problem

Consider the two-state Markov chain characterized by the transition matrix

$$P = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix},$$

with initial distribution

$$\pi_0 = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}.$$

The observable variable y_t can take the values

$$\bar{y} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}.$$

We compute the likelihood of the following observed histories for $t = 0, 1, \dots, 4$:

- a. 1, 5, 1, 5, 1
- b. 1, 1, 1, 1, 1
- c. 5, 5, 5, 5, 5

Relevant Theory

For a finite-state Markov chain with state history

$$x_0, x_1, \dots, x_T,$$

the probability of observing the complete path is

$$\Pr(x_0, x_1, \dots, x_T) = \pi_0(x_0) \prod_{t=0}^{T-1} P_{x_t, x_{t+1}}.$$

Thus, the likelihood of a particular history is obtained by multiplying

- the probability of the initial state,
- and the transition probabilities associated with each movement along the path.

In this exercise, the observable values uniquely identify the underlying states:

$$y_t = 1 \iff x_t = 0,$$

and

$$y_t = 5 \iff x_t = 1.$$

Analytical Solution

Part (a)

The observed history is

$$(1, 5, 1, 5, 1),$$

which corresponds to the state sequence

$$(0, 1, 0, 1, 0).$$

Its likelihood is therefore

$$\begin{aligned} L_a &= \Pr(X_0 = 0)P_{01}P_{10}P_{01}P_{10} \\ &= 0.5 \times 0.1 \times 0.3 \times 0.1 \times 0.3. \end{aligned}$$

Hence,

$$\boxed{L_a = 0.00045}$$

This history is relatively unlikely because it requires the chain to switch states at every period. In particular, the transition from state 0 to state 1 has probability only 0.1.

Part (b)

The observed history is

$$(1, 1, 1, 1, 1),$$

which corresponds to

$$(0, 0, 0, 0, 0).$$

The likelihood is

$$\begin{aligned} L_b &= \Pr(X_0 = 0)P_{00}^4 \\ &= 0.5(0.9)^4. \end{aligned}$$

Therefore,

$$\boxed{L_b = 0.32805}.$$

This path is relatively likely because state 0 is highly persistent:

$$P_{00} = 0.9.$$

Part (c)

The observed history is

$$(5, 5, 5, 5, 5),$$

corresponding to the state sequence

$$(1, 1, 1, 1, 1).$$

Its likelihood is

$$\begin{aligned} L_c &= \Pr(X_0 = 1)P_{11}^4 \\ &= 0.5(0.7)^4. \end{aligned}$$

Hence,

$$\boxed{L_c = 0.12005}.$$

State 1 is also persistent, but less so than state 0 because

$$P_{11} = 0.7 < P_{00} = 0.9.$$

Therefore,

$$\boxed{L_b > L_c > L_a}.$$

As we can see, the ranking reflects the transition matrix's persistence structure; indeed, remaining in state 0 is the most likely of the three histories, remaining in state 1 is less likely, and repeated switching between the two states is substantially less likely.

Log-Likelihood Representation

For longer histories, multiplying many probabilities can lead to numerical underflow. It is therefore often convenient to work with the log-likelihood. In more detail, the path likelihood

$$L = \pi_0(x_0) \prod_{t=0}^{T-1} P_{x_t, x_{t+1}}$$

can be rewritten as

$$\log L = \log \pi_0(x_0) + \sum_{t=0}^{T-1} \log P_{x_t, x_{t+1}}.$$

The original likelihood can then be recovered through

$$L = \exp(\log L),$$

which is a way more useful representation both in numerical and econometric applications.

Computational Extensions

The accompanying Python implementation extends the analytical solution in several directions.

1. Analytical and Computational Verification

The likelihoods obtained analytically are compared with those computed from the general log-likelihood function.

For each path,

$$L = \exp(\log L),$$

and the numerical results are checked against the analytical expressions

$$L_a = 0.5(0.1)(0.3)(0.1)(0.3),$$

$$L_b = 0.5(0.9)^4,$$

and

$$L_c = 0.5(0.7)^4.$$

The implementation verifies that the analytical and computational results coincide up to numerical precision.

2. Mapping Observed Values into Markov States

The exercise is expressed in terms of the observed values

$$y_t \in \{1, 5\},$$

whereas the Python implementation uses zero-based state indexing.

The correspondence is therefore

$$y_t = 1 \iff x_t = 0,$$

and

$$y_t = 5 \iff x_t = 1.$$

A general mapping function is implemented so that likelihoods can be computed directly from observed histories rather than requiring the user to manually convert observations into state indices.

3. Monte Carlo Simulation

A simulation function is used to generate finite histories from the Markov chain according to the transition matrix P .

The initial state can either be specified directly or drawn from the initial distribution

$$\pi_0 = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}.$$

Repeated simulation illustrates how the transition matrix governs the persistence and switching behavior of the process.

4. Monte Carlo Verification of Path Probabilities

The theoretical likelihood of each path is compared with its empirical frequency in a large number of simulated histories. For a sufficiently large number of simulations,

$$\widehat{\Pr}(x_0, \dots, x_T) \approx \Pr(x_0, \dots, x_T),$$

so the Monte Carlo experiment provides a numerical illustration of the relationship between theoretical probabilities and empirical frequencies.

In particular, the simulation should approximately reproduce the ordering

$$L_b > L_c > L_a.$$

This confirms that paths characterized by persistence in state 0 are substantially more common than paths requiring repeated switching between states.

Reference

Ljungqvist, L. and Sargent, T. J. (2018). *Recursive Macroeconomic Theory*, 4th edition, MIT Press.